

Software Requirements Specification (SRS)

Project: GrievanceConnect - Smart Hierarchical Student Grievance Management System

1. Introduction

GrievanceConnect is a web-based application that enables students to submit grievances either anonymously or with their identity. Complaints are routed to the appropriate authority and automatically escalated if unresolved within defined timelines.

2. Objectives

- Secure grievance submission
- Anonymous reporting
- Hierarchical escalation
- Transparent status tracking
- Role-based access control

3. Users

Student, Staff, HOD, Principal, Administrator

4. Functional Requirements

FR1 User registration/login
FR2 JWT authentication and role-based authorization
FR3 Complaint submission with attachments
FR4 Anonymous complaint option
FR5 Complaint status tracking
FR6 Complaint history
FR7 Automatic escalation
FR8 Notifications
FR9 Dashboard and reports
FR10 Feedback after resolution

5. Non-Functional Requirements

Security using Spring Security and BCrypt, performance with REST APIs, responsive UI, auditability, maintainability, and scalability.

6. Technology Stack

Frontend: React.js
Backend: Spring Boot (Gradle)
Database: MySQL
ORM: Spring Data JPA
Security: Spring Security + JWT

7. Database Tables

roles, departments, users, complaints, complaint_history, escalations, attachments, notifications, feedback

8. Future Enhancements

AI-based categorization, priority detection, sentiment analysis, duplicate complaint detection, email/SMS notifications.